Definition (Functorial stacking category)
A *functorial stacking category* is a stacking category where the two stacking operations $\otimes$ and $\otimes$ are the two components of a functor

$$\otimes : \mathbf{C} \times \mathbf{C} \to \mathbf{C}.$$

In infix notation, this means that, in addition to the equation

$$(f \fatsemi h) \otimes (g \fatsemi i) = (f \otimes g) \fatsemi (h \otimes i)$$

it is now also the case that $\otimes$ is compatible with identity morphisms:

$$\mathrm{id}_X \otimes \mathrm{id}_Y = \mathrm{id}_{X \otimes Y}$$

for all objects X, Y of $\mathbf{C}$.